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GitHub Link	https://github.com/192425245/ml0403

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA04 SLOT :

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I K.Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

Q1. CNN vs Fully Connected Neural Network

Scenario: Medical X-Ray Classification

A medical imaging organization needs to classify chest X-ray images. CNNs are compared with fully connected networks based on image structure, feature extraction, parameters and computation.

Parameter	CNN	Fully Connected Network
Architecture	Convolution + activation + pooling + dense	Mainly dense layers
Spatial features	Preserves local relationships	Flattening loses direct spatial structure
Parameters	Lower through local connections and shared filters	Very large for high-resolution images
Computation	Efficient for image patterns	Expensive for large images
Feature learning	Edges → textures → objects	Learns after flattened input
Image suitability	Highly suitable	Less suitable
Translation handling	Same filter can detect patterns at different locations	Needs separate learned connections

CNNs are preferred because images contain strong spatial relationships. Convolution extracts local patterns, pooling reduces dimensions, and parameter sharing avoids creating separate weights for every image location. This makes CNNs more efficient and better suited to medical image classification.

Q2. Different CNN Layers and Filters

Scenario: Industrial Product Defect Detection

A manufacturing company uses camera images to detect cracks, scratches and dents. Each CNN layer contributes a different stage of feature learning.

Layer	Function	Example
Convolution	Extracts local visual features	Edges and textures
ReLU	Adds non-linearity	Complex feature patterns
Pooling	Reduces spatial size	Compact feature maps
Fully Connected	Combines learned features	Final defect decision
Output	Produces prediction	Defect / No Defect

Filter Level	Features Learned	Defect Example
Early	Edges, corners, gradients	Crack boundary
Middle	Textures, curves, shapes	Scratch pattern
Deep	High-level combinations	Complex defect

Filter Levels in CNN for Industrial Defect Detection

Filter Level	Features Learned	Defect Example
Early Layers	Learn simple features such as edges, corners, lines, brightness changes, and gradients from the product image.	Detects the boundary of a crack , sharp edges, or small surface lines.
Middle Layers	Combine simple features to learn textures, curves, shapes, and repeated surface patterns .	Identifies scratch patterns, dents, rough textures, or irregular surfaces .
Deep Layers	Combine middle-level features to recognize complex and high-level defect patterns .	Detects a complete crack, major dent, broken component, or complex manufacturing defect .

Simple

explanation:

CNN filters learn features progressively. **Early filters detect basic patterns, middle filters identify textures and shapes, and deep filters recognize complete defect patterns.** This hierarchical learning helps the system automatically distinguish defective products from normal products without manually defining every defect.

Q3. Parameter Sharing vs Independent Weights

Scenario: Smart Surveillance at Large Scale

A surveillance system processes thousands of high-resolution images. Reducing memory and computation is important for efficient large-scale image processing.

Aspect	CNN Parameter Sharing	FCN Independent Weights
Weights	Same filter reused across locations	Separate weights for connections
Parameters	Much smaller	Can become extremely large
Memory	Lower	Higher
Computation	Efficient local operations	Expensive dense operations
Feature detection	Reusable visual detectors	More location-specific
Learning	Efficient reusable patterns	Requires many independent parameters
Large-scale use	Practical	Less practical

Parameter sharing allows one learned filter to detect the same edge or texture wherever it occurs. It greatly reduces storage and improves learning efficiency. It also provides useful translation-related robustness, although exact translation invariance depends on the architecture and training data.

Q4. AlexNet vs ResNet

Scenario: Advanced Image Classification Platform

A computer vision company is selecting an architecture for complex image recognition. AlexNet and ResNet are compared on depth, feature learning, optimization and computational requirements.

Criteria	AlexNet	ResNet
Architecture	Sequential CNN	Residual/skip-connected CNN
Depth	Classic model has 8 learned layers	Available from 18 to 152+ layers
Filters	Large early filters; conventional blocks	Mostly small convolution filters in modern variants
Feature learning	Strong historical baseline	Richer deep hierarchical features
Training	Less suited to very deep extensions	Skip connections ease deep training

Vanishing gradient	More concern as depth increases	Residual paths improve gradient flow
Complex recognition	Good baseline	Generally stronger
Modern use	Historical importance	Strong modern baseline

Classification

ResNet is recommended for complex modern recognition because residual connections make deep feature learning easier and reduce optimization problems associated with very deep networks. A lightweight ResNet variant can be selected when real-time computation is important.

Q5. Regularized CNN vs Non-Regularized CNN

Scenario: Plant Disease Classification

An agricultural organization trains a CNN to classify plant diseases from leaf images. The model has high training accuracy but poor unseen-image performance, indicating overfitting.

Aspect	Without Regularization	With Regularization
Training accuracy	May become extremely high	Usually more controlled
Validation performance	Often poor under overfitting	Usually improves
Generalization	Weak	Stronger
Model behaviour	May memorize training images	Learns robust patterns
Training stability	Can be sensitive	Batch normalization can stabilize training

Technique	Application	Benefit
Data Augmentation	Rotation, flip, crop, brightness changes	Handles real-world leaf variation
Dropout	Randomly deactivate neurons	Reduces feature co-adaptation
Batch Normalization	Normalize intermediate activations	Improves training stability
Early Stopping	Stop when validation stops improving	Limits over-training
L2 Regularization	Penalize large weights	Controls model complexity

Data augmentation is especially important because leaves can appear at different angles, scales and lighting conditions. Dropout and batch normalization can be added to improve generalization and training stability. The techniques should be balanced to avoid underfitting.

Overall Comparative Insight

Topic	Main Finding	Practical Decision
CNN vs FCN	CNN preserves spatial structure and shares weights	Use CNN for images
CNN Layers	Layers learn progressively richer features	Use hierarchical extraction
Parameter Sharing	Reduces parameters and memory	Essential for large image datasets
AlexNet vs ResNet	Residual connections support deeper learning	Prefer suitable ResNet
Regularization	Improves unseen-data performance	Use augmentation + suitable regularization

Conclusion

The comparisons show why CNNs are effective for real-world image processing. Convolution and parameter sharing make image learning efficient, pooling reduces spatial complexity, deeper filters learn meaningful patterns, ResNet enables deeper optimization through residual connections, and regularization improves generalization. Therefore, architecture selection should consider accuracy, computational resources, robustness and deployment requirements.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates	Provides evaluation	Limited evaluation;	No critical evaluation

		strengths, weaknesses, and implications with strong justification	with some justification	lacks depth	provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand